

STATISTICS



SYNOPSIS

FREQUENCY DISTRIBUTION

- **Class Limits:** The starting and end values of each class are called the lower limit and upper limit respectively of that class.

Ex. 1) The lower limit of the class 0-9 is 0

2) The upper limit of the class 50-59 is 59

- **Class boundaries :** The average of the upper limit of a class and the lower limit of the next class is called the upper boundary of that class. The upper boundary of a class becomes the lower boundary of the next class. These boundaries are called True class limits.

Ex. 1) 1-10, 11-20, 21-30 are the classes, the lower boundary of the class 11-20 is

$$\frac{10+11}{2} = 10.5$$

2) 60-69, 70-79, 80-89, 90-99 are the classes, the upper boundary of the class 70-79 is

$$\frac{79+80}{2} = 79.5$$

- **Class interval (or) the size of the class :** The difference between the lower limits or the upper limits of two consecutive classes is called the Class-interval (or) the size of the class.

Ex. The class interval in the frequency distribution with the classes 1-8, 9-16, 17-24 ... length of class = 9-1 = 8

- **Mid value of the class :** Mid value of class

$$1-10 \text{ is } \frac{1+10}{2} = 5.5$$

- For over lapping classes 0-10, 10-20, 20-30 etc

$$\text{the class mark of the class 0-10 is } \frac{0+10}{2} = 5$$

3) For non over lapping class 0-19, 20-39, 40-59,..... etc the class mark of the class 20-39 is

$$\left(\frac{20+39}{2} \right) = 29.5$$

- **Measures of Central Tendency:** One of the most important objectives of statistical analysis is to get one single value that describes the

characteristic of the entire data. Such a value is called the central value or an average.

The following are the important types of averages:

1. Arithmetic Mean
2. Geometric Mean
3. Harmonic mean
4. Median
5. Mode

We consider these measures in three cases (i) Individual series (i.e. each individual observation is given) (ii) discrete series (i.e. the observations along with number of times a particular observation called the frequency is given) (iii) continuous series (i.e. the class intervals along with their frequencies are given)

Arithmetic Mean

- **Individual Series :** If $x_1, x_2, \dots, x_n$ are the values of the variable x , then the arithmetic mean usually denoted by $\bar{x}$ or μ or $E(x)$ is given by

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{1}{n} \sum_{i=1}^n x_i$$

Note: A.M. $(\bar{x}) = A + \frac{\sum (x_i - A)}{n}$ where A is the assumed average. (For individual series)

- **Discrete Series :** If a variable takes values $x_1, x_2, \dots, x_n$ with corresponding frequencies $f_1, f_2, \dots, f_n$ then the arithmetic mean $\bar{x}$ is given by

$$\bar{x} = \frac{f_1 x_1 + f_2 x_2 + \dots + f_n x_n}{f_1 + f_2 + \dots + f_n} = \frac{1}{N} \sum_{i=1}^n f_i x_i,$$

$$\text{where } N = \sum_{i=1}^n f_i$$

- **Continuous Series :** In case of a set of data with class intervals, we cannot find the exact value of the mean because we do not know the exact values of the variables. We, therefore, try to obtain an approximate value of the mean. The method of approximate is to replace all the observed values belonging to a class by mid-value of the class. If $x_1, x_2, \dots, x_n$ are the mid values of the class intervals having corresponding frequencies $f_1, f_2, \dots, f_n$ then we apply the same formula as in discrete series.

$$\bar{x} = \frac{1}{N} \sum_{i=1}^n f_i x_i, \quad N = \sum_{i=1}^n f_i$$

- **Combined Arithmetic Mean:** If $\bar{x}_i (i = 1, 2, \dots, k)$ are the means of k - series of sizes $n_i (i = 1, 2, 3, \dots, k)$ respectively, then the combined or composite mean $\bar{x}$ can be obtained by the formula :

$$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2 + \dots + n_k\bar{x}_k}{n_1 + n_2 + \dots + n_k} = \frac{\sum n_i \bar{x}_i}{\sum n_i}$$

- **Weighted Arithmetic Mean:**

Let $w_1, w_2, \dots, w_n$ be the weights assigned to the values $x_1, x_2, \dots, x_n$ respectively of a variable x ,

then the weighted A.M. is $\bar{x} = \frac{\sum w_i x_i}{\sum w_i}$.

Properties of Arithmetic Mean

- Sum of all the deviations from arithmetic mean is zero i.e.,

$$\sum_{i=1}^n (x_i - \bar{x}) = 0 \text{ (in case of individual series)}$$

$$\sum_{i=1}^n f_i (x_i - \bar{x}) = 0 \text{ (in case of discrete or continuous series)}$$

- If each observation is increased or decreased by a given constant K , the mean is also increased or decreased by K

The property is also known as effect of change of origin. K can be taken to be any number. However, to simplify the calculations, K should be taken as a value which is in the middle of the table.

- Step Deviation Method or change of scale

If $x_1, x_2, \dots, x_n$ are mid values of class intervals with corresponding frequencies $f_1, f_2, \dots, f_n$ then

we may change the scale by taking $d_i = \frac{x_i - A}{h}$, in this case.

$$\bar{x} = A + h \times \left(\frac{1}{N} \sum f_i d_i \right) \text{ (if } A \text{ is assumed mean)}$$

A and h can be any numbers but if the lengths of class intervals are equal then h may be taken as width of the class interval.

In particular if each observation is multiplied or divided by a constant, the mean is also multiplied or divided by the same constant.

- The sum of the squared deviation of the variate from their mean is minimum i.e., the quantity $\sum (x_i - A)^2$ or $\sum f_i (x_i - A)^2$ is minimum when $A = \bar{x}$
- $E(aX + b) = aE(X) + b$ (where $E(X)$ = Mean of X)

W.E-1: The weighted mean of the first n natural numbers, the weights being the corresponding numbers, is

Sol. First n natural numbers are $1, 2, 3, \dots, n$; whose corresponding weights are $1, 2, 3, \dots, n$ respectively.

$$\begin{aligned} \therefore \text{weight mean} &= \frac{1 \times 1 + 2 \times 2 + \dots + n \times n}{1 + 2 + \dots + n} \\ &= \frac{1^2 + 2^2 + \dots + n^2}{1 + 2 + \dots + n} \\ &= \frac{n(n+1)(2n+1)}{6n(n+1)} = \frac{2n+1}{3} \end{aligned}$$

W.E-2: The weighted mean of the first n natural numbers whose weights are equal to the squares of the corresponding numbers is

$$\begin{aligned} \text{Sol. weighted mean} &= \frac{1 \cdot 1^2 + 2 \cdot 2^2 + \dots + n \cdot n^2}{1^2 + 2^2 + \dots + n^2} \\ &= \frac{\sum n^3}{\sum n^2} = \frac{\frac{n(n+1)}{2} \cdot \frac{n(n+1)}{2}}{\frac{n(n+1)(2n+1)}{6}} = \frac{3n(n+1)}{2(2n+1)} \end{aligned}$$

W.E-3: The average salary of male employees in a firm is Rs. 5200 and that of females is Rs.4200. The mean salary of all the employees is Rs.5000. The percentage of male and female employees are respectively is

Sol. Let $x_1 = 5200, x_2 = 4200, \bar{x} = 5000$

$$\text{Also, we know that } \bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$$

$$\Rightarrow 5000(n_1 + n_2) = 5200n_1 + 4200n_2 \Rightarrow \frac{n_1}{n_2} = \frac{4}{1}$$

$\therefore$ The percentage of male employees in the firm

$$= \frac{4}{4+1} \times 100 = 80\%$$

and the percentage of female employees in the firm

$$= \frac{1}{4+1} \times 100 = 20\%$$

W.E-4: If the mean of 9 observations is 100 and mean of 6 observations is 80, then the mean of 15 observations is

Sol. $n_1 = 9, \bar{x}_1 = 100$ and $n_2 = 6, \bar{x}_2 = 80$

$$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2} = \frac{9 \times 100 + 6 \times 80}{9 + 6} = 92$$

W.E-5: If a variate X is expressed as a linear function of two variates U and V in the form $X = aU + bV$ then the mean $\bar{X}$ of X is

Sol. we have $\Sigma X = a\Sigma U + b\Sigma V$

$$\bar{X} = \frac{1}{n} \Sigma X = a \cdot \frac{1}{n} \Sigma U + b \cdot \frac{1}{n} \Sigma V$$

$$\Rightarrow \bar{X} = a\bar{U} + b\bar{V}$$

W.E-6: If the arithmetic mean of the numbers $x_1, x_2, x_3, \dots, x_n$ is $\bar{x}$, then the arithmetic mean of the numbers

$ax_1 + b, ax_2 + b, ax_3 + b, \dots, ax_n + b$, where a, b are two constants, would be

Sol. Required mean

$$= \frac{(ax_1 + b) + (ax_2 + b) + \dots + (ax_n + b)}{n}$$

$$= \frac{a(x_1 + x_2 + \dots + x_n)}{n} + b = a\bar{x} + b$$

$$\left(\because \frac{x_1 + x_2 + \dots + x_n}{n} = \bar{x} \right)$$

W.E-7: If the mean of the numbers $27 + x, 31 + x, 89 + x, 107 + x, 156 + x$ is 82, then the mean of $130 + x, 126 + x, 68 + x, 50 + x, 1 + x$ is

Sol. Given

$$82 = \frac{(27 + x) + (31 + x) + (89 + x) + (107 + x) + (156 + x)}{5}$$

$$\Rightarrow 82 \times 5 = 410 + 5x \Rightarrow 410 - 410 = 5x \Rightarrow x = 0$$

Therefore, the required mean is

$$\bar{x} = \frac{130 + x + 126 + x + 68 + x + 50 + x + 1 + x}{5}$$

$$= \frac{375 + 5x}{5} = 75$$

W.E-8: A student obtain 75%, 80% and 85% in three subjects. If the marks of another subject is added, then his average cannot be less than

Sol. Marks obtained from three subjects out of 300 is

$$75 + 80 + 85 = 240$$

If the marks of another subject is added, then the marks will be ≥ 240 out of 400

$$\therefore \text{minimum average marks} = \frac{240}{4} = 60\%$$

[when marks in the fourth subject = 0]

W.E-9: The mean of 100 items is 49. It was discovered that three items which should have been 60, 70, 80 were wrongly read as 40, 20, 50 respectively. The correct mean is

Sol. Sum of 100 items = $49 \times 100 = 4900$

$$\text{sum of items added} = 60 + 70 + 80 = 210$$

$$\text{new sum} = 4900 + 210 - 110 = 5000$$

$$\therefore \text{correct mean} = \frac{5000}{100} = 50$$

W.E-10: The mean weight per student in a group of seven students is 55kg. If the individual weights of six students are 52, 58, 55, 53, 56 and 54, then the weight of the seventh student is

Sol. The total weight of seven students is $55 \times 7 = 385\text{kg}$

The sum of the weights of six students is

$$52 + 58 + 55 + 53 + 56 + 54 = 328\text{kg}$$

Hence, the weight of the seventh student is

$$= 385 - 328 = 57\text{kg}$$

Geometric Mean

➤ In case of individual series $x_1, x_2, \dots, x_n$

$$\text{G.M.} = (x_1 x_2 \dots x_n)^{1/n}$$

In case of discrete or continuous series

$$\text{G.M.} = (x_1^{f_1} x_2^{f_2} \dots x_n^{f_n})^{1/N}, \text{ where } N = \sum_{i=1}^n f_i$$

W.E-11: The geometric mean of the numbers

$$3, 3^2, 3^3, \dots, 3^n \text{ is}$$

Sol. $\therefore \text{G.M.} = (3 \cdot 3^2 \cdot \dots \cdot 3^n)^{1/n}$

$$= 3^{\frac{1+2+\dots+n}{n}} = 3^{\frac{n(n+1)}{2n}} = 3^{\frac{n+1}{2}}$$

- **Harmonic Mean:** The harmonic mean is based on the reciprocals of the value of the variable

$$\text{H.M.} = \frac{1}{\frac{1}{n} \left(\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n} \right)} \text{ or } \frac{1}{H} = \frac{1}{n} \sum_{i=1}^n \frac{1}{x_i}$$

(In case of Individual series)

$$\text{and } \frac{1}{H} = \frac{1}{N} \sum_{i=1}^n f_i \frac{1}{x_i} \text{ (in case of discrete series or continuous series)}$$

If $x_1, x_2, \dots, x_n > 0$ then it is known that

$$\text{A.M} \geq \text{G.M} \geq \text{H.M}$$

W.E-12: Find the harmonic mean of

$\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots, \frac{n}{n+1}$, occurring with frequencies 1, 2, 3, ..., n, respectively.

Sol. We know that, Harmonic mean $= \frac{\sum f}{\sum \left(\frac{f}{x} \right)}$

$$\sum f = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

$$\text{and } \sum \frac{f}{x} = \frac{1}{1/2} + \frac{2}{2/3} + \frac{3}{3/4} + \dots + \frac{n}{n/(n+1)}$$

$$= 2 + \frac{3 \times 2}{2} + \frac{4 \times 3}{3} + \dots + \frac{n(n+1)}{n}$$

$$= 2 + 3 + 4 + \dots + n + (n+1)$$

Which is an arithmetic progression with $a = 2$ and $d = 1$.

By the formula of sum of n term of an A.P,

$$\sum \left(\frac{f}{x} \right) = \left[\frac{n}{2} \{ 2a + (n-1)d \} \right]$$

$$\text{we have } = \frac{n}{2} \{ 2 \times 2 + n - 1 \} = \frac{n}{2} (3 + n)$$

∴ Harmonic mean

$$= \frac{2}{n(3+n)} = \frac{n(n+1) \times 2}{n(3+n) \times 2} = \frac{n+1}{3+n}$$

Median

- **Individual Series :** If the items are arranged in ascending or descending order of magnitude then the middle value is called median.

In case of odd number of values Median = size of $\frac{n+1}{2}$ th item. In case of even number of values

Median = average of $\frac{n}{2}$ th and $\frac{n+2}{2}$ th observation.

- **Discrete Frequency Distribution :** Arrange the data in ascending or descending order. Find the cumulative frequencies.

Apply the formula :

Median = Size of $\left(\frac{N+1}{2} \right)$ th item (N is odd)

$$= \frac{1}{2} \left[\left(\frac{N}{2} \right)^{\text{th}} \text{ observation} + \left(\frac{N}{2} + 1 \right)^{\text{th}} \text{ observation} \right] \quad (\text{N is even})$$

$$N = \sum f_i = \text{sum of frequencies}$$

- **Continuous Frequency Distribution :**

Consider the cumulative frequency (c.f.). Find $\frac{N}{2}$,

where $N = \sum_{i=1}^n f_i$. Find the cumulative frequency

(c.f.) just more than $N/2$. The corresponding value of x is median. In case of continuous distribution,

the class corresponding to c.f. just more than $\frac{N}{2}$ is

called the median class and the median is obtained

$$\text{by Median} = l + \frac{h}{f} \left(\frac{N}{2} - C \right)$$

Where l = the lower limit of the median class;

f = the frequency of the median class;

h = the width of the median class;

C = the c.f. of the class preceding to the median class and

$$N = \sum_{i=1}^n f_i$$

W.E-13: The median of a set of nine distinct observations is 20.5. If each of the last four observations of the set is increased by 2, then the median of the new set is

Sol. Since $n = 9$, median term = $\left(\frac{9+1}{2}\right)^{th} = 5^{th}$ term.

Now, the last four observations are increased by 2. Since the median is the 5th observation, which remains unchanged, there will be no change in median.

W.E-14: If a variable takes the discrete value

$$\alpha - 4, \alpha - \frac{7}{2}, \alpha - \frac{5}{2}, \alpha - 3, \alpha - 2, \alpha + \frac{1}{2},$$

$$\alpha - \frac{1}{2}, \alpha + 5 (a > 0), \text{ then the median is}$$

Sol. Arrange the data as follows:

$$\alpha - \frac{7}{2}, \alpha - 3, \alpha - \frac{5}{2}, \alpha - 2, \alpha - \frac{1}{2},$$

$$\alpha + \frac{1}{2}, \alpha - 4, \alpha + 5$$

$$\text{median} = \frac{1}{2} [\text{value of 4th item} + \text{value of 5th item}]$$

$$\therefore \text{median} = \frac{\alpha - 2 + \alpha - \frac{1}{2}}{2} = \alpha - \frac{5}{4}$$

W.E-15: The median of distribution 83, 54, 78, 64, 90, 59, 67, 72, 70, 73 is

Sol. On arranging in ascending order, we get 54, 59, 64, 67, 70, 72, 73, 78, 83, 90

$$n = 10$$

$$\therefore \text{median} = \frac{\text{value of } \frac{10}{2} \text{ term} + \text{value of } \left(\frac{10}{2} + 1\right)^{th} \text{ term}}{2}$$

$$= \frac{\text{value of 5th term} + \text{value of 6th term}}{2}$$

$$= \frac{70 + 72}{2} = 71$$

Mode : The mode is that value in a series of observations which occurs with greatest frequency. In case of **individual series**, the mode is the value which occurs more frequently

In case of **discrete series**, quite often mode can be determined just by inspection i.e. by looking to that value of variable around which the items are most heavily concentrated.

In case of **continuous series**,

$$\text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_2 - f_0} \times h$$

Where l = the lower limit of the modal class i.e. the class having maximum frequency;
 f_1 = frequency of the modal class;
 f_0 = frequency of the class preceding the modal class;
 f_2 = frequency of the class succeeding the modal class and

h = width of the modal class.

- Relation between Mean, Median and Mode is
 mean - mode = 3 (mean - median) or
 Mode = 3 median - 2 mean

W.E-16: The mode of the following distribution is

Class interval	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Frequency	5	8	7	12	28	20	10	10

Sol. Here, maximum frequency is 28. Thus, the class 40-50 is the modal class.

$$\begin{aligned} \text{Mode} &= l + \frac{f_1 - f_0}{2f_1 - f_2 - f_0} \times h \\ &= 40 + \frac{10(28 - 12)}{(2 \times 28 - 12 - 20)} \\ &= 40 + 6.666 = 46.67 (\text{approx.}) \end{aligned}$$

W.E-17: If in a frequency distribution, the mean and median are 20 and 21 respectively, then its mode is approximately

Sol. mode = 3 median - 2 mean = 3 (21) - 2(20) = 23

W.E-18: If in a moderately asymmetrical distribution the mode and the mean of the data are 6λ and 9λ , respectively, then the median is

Sol. For a moderately skewed distribution,
 mode = 3 median - 2 mean

$$\Rightarrow 6\lambda = 3\text{median} - 18\lambda \Rightarrow \text{median} = 8\lambda$$

Measures of Dispersion: Literally, dispersion means 'scatteredness'. Dispersion measures the degree of scatteredness of the variable about a central value. Different measures of dispersion are

1. Range
2. Mean-deviation
3. Quartile deviation
4. Standard deviation

➤ **Range:** The range is the difference between the largest and smallest observation.

$$\text{Coefficient of Range} = \frac{\text{Range}}{\text{Maximum} + \text{Minimum}}$$

➤ **Mean-deviation:** If $x_1, x_2, \dots, x_n$ are n observations then mean deviation about a point M is given by

$$\text{M.D.} = \frac{1}{n} \sum |x_i - M| \text{ where } M \text{ is mean or median or mode}$$

In case of discrete or continuous series

$$\text{M.D.} = \frac{1}{N} \sum f_i |x_i - M|, N = \sum f_i$$

M.D. is least when taken from the median

$$\text{Coefficient of Mean Deviation} = \frac{\text{Mean Deviation}}{M}$$

where M is the Mean, Median or Mode

➤ **Quartile Deviation:** $Q.D. = \frac{Q_3 - Q_1}{2}$, where

Q_3 and Q_1 , are the third quartile and the first quartile. Q_1 and Q_3 can be calculated in a similar manner as median. In fact, quartiles divides the data into four parts.

In case of **individual series**, arrange the data in ascending or descending order.

$$Q_1 = \text{size of } \frac{n+1}{4} \text{th and } Q_3 = \text{size of } \frac{3(n+1)}{4} \text{th item}$$

In case of **discrete frequency distribution**, Q_1 is obtained by considering cumulative frequency. Find $N/4$, where $N = \sum f_i$. Find the cumulative frequency (c.f.) just more than $N/4$. The corresponding value of x is Q_1 . Similarly for obtaining Q_3 , find $3N/4$ and the c.f. just more than $3N/4$. The corresponding value of x is Q_3 . In case of continuous distribution.

$$Q_i = l + \frac{h}{f} \left(i \frac{N}{4} - c \right), i = 1, 2, 3$$

Where l = the lower limit of the class whose c.f. is just more than $iN/4$,

f is its frequency and h is its width. C = c.f. of the class preceding to the class whose c.f. is just more than $iN/4$, $i = 1, 2, 3$.

Note that $i = 2$ will give us median.

$$\text{Coefficient of Quartile Deviation} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

W.E-19: The quartile deviation of daily wages of in (Rs.) of 11 persons given below 140, 145, 130, 165, 160, 125, 150, 170, 175, 120, 180

Sol. The given data in ascending order of magnitude is 120, 125, 130, 140, 145, 150, 160, 165, 170, 175, 180

$$\text{Here, } Q_1 = \left(\frac{n+1}{4} \right)^{\text{th}} \text{ term} = \frac{11+1}{4} \text{ term} = 3^{\text{rd}} \text{ term} = 130$$

$$Q_3 = \frac{3(n+1)}{4} = \frac{3 \times (11+1)}{4} = 9^{\text{th}} = 170$$

$$QD = \frac{Q_3 - Q_1}{2} = \frac{170 - 130}{2} = \frac{40}{2} = 20$$

➤ **Standard Deviation:** Variance σ^2 in case of **individual series** is given by

$$\sigma^2 = \frac{1}{n} \sum (x_i - \bar{x})^2 = \frac{1}{n} \sum x_i^2 - \left(\frac{1}{n} \sum x_i \right)^2$$

If $x_1, x_2, \dots, x_n$ occur with frequency $f_1, f_2, \dots, f_n$ respectively then σ^2 (variance)

$$= \frac{1}{N} \sum (x_i - \bar{x})^2 = \frac{1}{N} \sum f_i x_i^2 - \left(\frac{1}{N} \sum f_i x_i \right)^2$$

Standard deviation is the positive square root of variance

There is no effect of change of origin on standard deviation

$$\sigma_x^2 = h^2 \left[\frac{1}{N} \sum f_i d_i^2 - \left(\frac{1}{N} \sum f_i d_i \right)^2 \right]$$

Coefficient of Standard Deviation is $\frac{\sigma}{\bar{x}}$.

$$\text{Coefficient of variation} = \frac{\sigma}{\bar{x}} \times 100$$

- **Combined variance:** If there are two samples of sizes n_1 and n_2 with $\bar{x}_1$ and $\bar{x}_2$ as their means σ_1 and σ_2 their standard deviations respectively, then the combined variance is given by

$$\sigma^2 = \frac{1}{n_1 + n_2} \left[n_1 \sigma_1^2 + n_2 \sigma_2^2 + \frac{n_1 n_2}{n_1 + n_2} (\bar{x}_1 - \bar{x}_2)^2 \right]$$

$$\text{or } \sigma^2 = \frac{n_1 (\sigma_1^2 + d_1^2) + n_2 (\sigma_2^2 + d_2^2)}{n_1 + n_2}$$

where $d_1 = \bar{x}_1 - \bar{x}$ and $d_2 = \bar{x}_2 - \bar{x}$, $\bar{x}$ being the combined mean.

Note :1

if $V(X)$ is variance of X then

$$V(X + a) = V(X)$$

$$V(aX) = a^2 V(X)$$

$$V(aX + b) = a^2 V(X)$$

$$V(aX + bY) = a^2 V(X) + b^2 V(Y)$$

Note :2

For $a, a + d, a + 2d, \dots, a + (n - 1)d$,

$$\bar{x} = a + \frac{(n-1)d}{2}; \sigma^2 = \frac{n^2 - 1}{12} d^2$$

Note:3

$$1) Q.D < M.D < S.D$$

$$2) \frac{Q.D}{10} = \frac{M.D}{12} = \frac{S.D}{15}$$

W.E-20: The variance of the first 'n' natural numbers is

Sol. Variance

$$\begin{aligned} &= (SD)^2 = \frac{1}{n} \sum x^2 - \left(\frac{\sum x}{n} \right)^2, \left(\because \bar{x} = \frac{\sum x}{n} \right) \\ &= \frac{n(n+1)(2n+1)}{6n} - \left(\frac{n(n+1)}{2n} \right)^2 = \frac{n^2 - 1}{12} \end{aligned}$$

W.E-21: If the M.D is 12, the value of S.D will be

Sol. We know that $Q.D = \frac{5}{6} \times M.D = \frac{5}{6} \times 12 = 10$

$$\therefore S.D = \frac{3}{2} \times Q.D = \frac{3}{2} \times 10 = 15$$

W.E-22: The mean of five observations is 4 and their variance is 5.2. If three of these observations are 1, 2 and 6, then the other two observations are

Sol. Let the two unknown items be x and y , then

$$\text{Mean} = 4 \Rightarrow \frac{1 + 2 + 6 + x + y}{5} = 4$$

$$\Rightarrow x + y = 11 \dots (1) \text{ and variance} = 5.2$$

$$\Rightarrow \frac{1^2 + 2^2 + 6^2 + x^2 + y^2}{5} - (\text{mean})^2 = 5.2$$

$$41 + x^2 + y^2 = 5[5.2 + (4)^2]$$

$$41 + x^2 + y^2 = 106$$

$$x^2 + y^2 = 65 \dots (2)$$

Solving (1) and (2) for x and y , we get

$x = 4, y = 7$ or $x = 7, y = 4$.



C.U.Q

1. The arithmetic mean of the series

${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_n$ is

$$1) \frac{2^n}{(n+1)} \quad 2) \frac{2^n}{n} \quad 3) \frac{2^{n-1}}{(n+1)} \quad 4) \frac{2^n}{(n-1)}$$

2. The arithmetic mean of the squares of the first n natural numbers is

$$\begin{aligned} 1) \frac{(n+1)}{6} \quad 2) \frac{(n+1)(2n+1)}{6} \\ 3) \frac{(n^2-1)}{6} \quad 4) n \end{aligned}$$

3. The standard deviation of a variable x is σ .

The standard deviation of the variable $\frac{ax+b}{c}$

where a, b, c are constants is ...

$$1) \left(\frac{a}{c} \right) \sigma \quad 2) \left| \frac{a}{c} \right| \sigma \quad 3) \left| \frac{a^2}{c^2} \right| \sigma \quad 4) \left(\frac{2a}{c} \right) \sigma$$

4. If the mean of the set of numbers $x_1, x_2, \dots, x_n$ is $\bar{x}$, then the mean of the numbers

$x_i + 2i, 1 \leq i \leq n$ is

$$1) \bar{x} + n + 1 \quad 2) \bar{x} + 2n \quad 3) \bar{x} + 2 \quad 4) \bar{x} + n$$

5. If each observation of a raw data, whose variance is σ^2 , is multiplied by λ , then the variance of the new set is
 1) σ^2 2) $\sigma^2\lambda^2$ 3) $\sigma^2 + \lambda$ 4) $\sigma^2 + \lambda^2$
6. If a variable takes values 0, 1, 2, 3, with frequencies proportional to $e^{-\lambda}, e^{-\lambda}\lambda, \frac{e^{-\lambda}\lambda^2}{2!}, \frac{e^{-\lambda}\lambda^3}{3!}, \dots$ then the mean of the distribution is
 1) $e^{-\lambda}$ 2) λ 3) $e^{-\lambda}\lambda$ 4) $\left(\frac{1}{2}\right)e^{-\lambda}\lambda^2$
7. A cyclist covers his first three miles at an average speed of 8 m.p.h. Another two miles at 3 m.p.h. and the last two miles at 2 m.p.h. The average speed for the entire journey is : (in m.p.h.)
 1) 3 2) 2.4 3) 3.8 4) 3.43

C.U.Q - KEY

- 1) 1 2) 2 3) 2 4) 1 5) 2 6) 2 7) 4

C.U.Q - HINTS

1. $\frac{n_{c_0} + n_{c_1} + \dots + n_{c_n}}{n+1} = \frac{2^n}{n+1}$
2. $\frac{n(n+1)(2n+1)}{6n} = \frac{(n+1)(2n+1)}{6}$
3. $S.D(ax+b) = |a|S.D(x)$
4. $\frac{x_1 + 2 + x_2 + 4 + x_3 + 6 + \dots + x_n + 2n}{n}$
 $= \frac{\sum x_i}{n} + \frac{2+4+6+\dots+2n}{n} = \bar{x} + (n+1)$
5. $\sigma^2\lambda^2$
6. Mean $= \frac{e^{-\lambda}\left(\lambda + \frac{2\lambda^2}{2} + \frac{3\lambda^3}{3} + \frac{4\lambda^4}{4} + \dots\right)}{1}$
 $= \lambda e^{-\lambda}\left(1 + \frac{\lambda}{1} + \frac{\lambda^2}{2} + \frac{\lambda^3}{3} + \dots\right)$
 Mean $= \lambda e^{-\lambda}(e^\lambda) = \lambda$
7. $\frac{\text{total distance}}{\text{total time}}$



LEVEL-I (C.W)

1. For overlapping classes 0-10, 10-20, 20-30, etc. Then the class mark of the class 0-10 is
 1) 0 2) 10 3) 5 4) 6
2. Which one of the following measures is the most suitable one of central location for computing intelligence of students?
 1) Mode 2) A.M. 3) G.M. 4) Median

MEAN (A.M, G.M, H.M)

3. The mean of 20 observations is 15. On checking it was found that two observations were wrongly copied as 3 and 6. If wrong observations are replaced by correct values 8 and 4, then the correct mean is
 1) 15 2) 15.15 3) 16.15 4) 17
4. The mean weight of 9 items is 15. If one more item is added to the series the mean becomes 16. The value of 10th item is
 1) 35 2) 30 3) 25 4) 20
5. When 15 was subtracted from each of the seven observations the following number resulted : -3, 0, -2, 4, 6, 1, 1. The mean of the distribution is
 1) 14 2) 15 3) 16 4) 17
6. Mean of 100 items is 49. It was discovered that three items which should have been 60, 70, 80 were wrongly read as 40, 20, 50 respectively. The correct mean is.

- 1) 48 2) $82\frac{1}{2}$ 3) 80 4) 50

7. If the arithmetic and harmonic means of two numbers are 4.5 and 4 respectively, then one of the number is
 1) 5 2) 6 3) 7 4) 4

MEDIAN & MODE

8. If the mode of a data is 18 and the mean is 24, then median is
 1) 18 2) 24 3) 21 4) 22
9. If the median of 21 observations is 40 and if the observations greater than the median are increased by 6 then the median of the new data will be
 1) 40 2) 46 3) $46 + \frac{40}{21}$ 4) $46 - \frac{40}{21}$

10. Mode of the data 3, 2, 5, 2, 3, 5, 6, 6, 5, 3, 5, 2, 5 is
 1) 6 2) 4 3) 3 4) 5

11. Mode of the distribution

Marks	4	5	6	7	8
No. of students	3	5	10	6	1
1) 6	2) 10	3) 8	4) 4		

RANGE, Q.D, S.D AND VARIANCE

12. The range of the following set of observations 2, 3, 5, 9, 8, 7, 6, 5, 7, 4, 3 is
 1) 11 2) 7 3) 5.5 4) 6
13. The quartile deviation of daily wages (in Rs.) of 7 persons given below is 12, 7, 15, 10, 17, 17, 25 is
 1) 14.5 2) 5 3) 3.5 4) 4.5
14. If the standard deviation of 0, 1, 2, 3, ..., 9 is K, then the standard deviation of 10, 11, 12, 13, ..., 19 is
 1) K + 10 2) K 3) $\sqrt{10} + K$ 4) 10 K
15. The variance of the first n natural numbers is
 1) $\frac{n^2-1}{12}$ 2) $\frac{n^2-1}{6}$ 3) $\frac{n^2+1}{6}$ 4) $\frac{n^2+1}{12}$
16. The mean of four observations is 3. If the sum of the squares of these observations is 48 then their standard deviation is [EAMCET-2014]
 1) $\sqrt{2}$ 2) $\sqrt{3}$ 3) $\sqrt{5}$ 4) $\sqrt{7}$
17. If $x_1, x_2, \dots, x_n$ are n observations such that

$\sum_{i=1}^n x_i^2 = 400$ and $\sum_{i=1}^n x_i = 80$ then the least value of n is [EAMCET-2014]

- 1) 12 2) 15 3) 16 4) 18
18. The sum of 10 items is 12 and sum of their squares is 18, then standard deviation is
 1) $-3/5$ 2) $6/5$ 3) $4/5$ 4) $3/5$
19. The mean of two samples of sizes 200 and 300 were found to be 25, 10 respectively. Their standard deviations were 3 and 4 respectively. The variance of combined sample of size 500 is
 1) 64 2) 65.2 3) 67.2 4) 64.2

LEVEL-I (C.W) - KEY

- 1) 3 2) 4 3) 2 4) 3 5) 3 6) 4 7) 2
 8) 4 9) 1 10) 4 11) 1 12) 2 13) 3 14) 2
 15) 1 16) 2 17) 3 18) 4 19) 3

LEVEL-I (C.W) - HINTS

1. 5 (mid value of the class)
3. $\frac{20 \times 15 - 3 - 6 + 8 + 4}{20}$
4. $\frac{9.15 + x}{10} = 16 \Rightarrow 135 + x = 160 \Rightarrow x = 25$
5. Mean = $\frac{-3 + 0 - 2 + 4 + 6 + 1}{7} = 1$
 The mean of the original distribution = $1 + 15 = 16$
6. $\frac{4900 - 40 - 20 - 50 + 60 + 70 + 80}{100} = \frac{5000}{100} = 50$
7. $\frac{a+b}{2} = 4.5$, $\frac{2ab}{a+b} = 4$, $a+b = 9$,
 $ab = 18 \Rightarrow a = 6$
8. Mode = 3 median - 2 mean, $18 = 3(\text{median}) - 2$
 (24) , Median = $\frac{66}{3} = 22$
9. 40
10. 3, 3, 3, 2, 2, 2, 5, 5, 5, 5, 5, 6, 6 mode = 5
11. The number having maximum frequency
12. $9 - 2 = 7$
13. $\frac{Q_3 - Q_1}{2}$ where $Q_1 = 10$, $Q_3 = 17$
14. K
15. $\frac{n^2 - 1}{12}$
16. $\bar{x} = 3$, $\sum x_i^2 = 48$; $\sigma^2 = \frac{1}{n} \sum x_i^2 - (\bar{x})^2$
 $= \frac{1}{4} \times 48 - 9 = 12 - 9 = 3$; $\sigma = \sqrt{3}$
17. $\sigma^2 \geq 0$, $\frac{1}{n} \sum x_i^2 - (\bar{x})^2 \geq 0$
18. $\sum x_i = 12$, $\sum x_i^2 = 18$, $n = 10$
 $S.D = \sqrt{\frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2}$
19. Combined mean $\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$; $d_1 = \bar{x}_1 - \bar{x}$,
 $d_2 = \bar{x}_2 - \bar{x}$ and use the formula Variance of
 combined data = $\sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{(n_1 + n_2)}$



LEVEL-I (H.W)

- For the class intervals 0-5, 5-10, 10-15, etc. The length of the class is
1) 5 2) 10 3) 0 4) 6
- Which of the following is not a measure of dispersion?
1) Variance 2) Mean Deviation
3) Mode 4) Standard Deviation

MEAN (A.M, GM, H.M)

- Which of the following would you regard as discrete variable
1) height 2) weight
3) time 4) number of persons in family
- The average salary of male employees in a firm was Rs. 520 and that of females was Rs. 420. The mean salary of all the employees was Rs. 500. The percentage of male and female employees are
1) 30, 70 2) 80, 20 3) 40, 60 4) 50, 50
- Mean of 10 numbers is 6. It was later observed that one number was misread as 9. When the correct mean was 7, then the correct value of that number is
1) 19 2) 20 3) 8 4) 10

MEDIAN & MODE

- The median of the scores 25, 28, 20, 8, 10, 15 is
1. 17.5 2. 16.5 3. 20.2 4. 13.5
- A data consists of two 2's, four 4's, six 6's, three 8's, and 10. Then the mode of data is
1) 2 2) 4 3) 6 4) 8
- In a moderately asymmetrical series, the values of arithmetic mean and mode are at 20.6 and 34.1 respectively. The value of the median is
1) 25.1 2) 28.0 3) 23.4 4) 35.3

RANGE, Q.D, S.D AND VARIANCE

- For a symmetric distribution $Q_1=20$, $Q_3=40$ the median of data is
1) 20 2) 30 3) 40 4) 10
- The standard deviation of the data given by
Variate (x) 0 1 2 3..... n

Frequency (f) nC_0 nC_1 nC_2 nC_3 nC_n

- $\sqrt{\frac{(n+1)}{2}}$ 2) $\sqrt{\frac{n}{2}}$ 3) $\frac{2^n}{n}$ 4) $\frac{\sqrt{n}}{2}$

- Variance of the data 2, 4, 6, 8, 10 is

1) 6 2) 7 3) 8 4) 9

LEVEL-I (H.W) - KEY

- 1) 1 2) 3 3) 4 4) 2 5) 1 6) 1 7) 3
8) 1 9) 2 10) 4 11) 3

LEVEL-I (H.W) - HINTS

- 5
- Mode
- Number of persons in family
- $\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$
- $\frac{60 - 9 + x}{10} = 7$
- 25, 28, 20, 8, 10, 15 → 8, 10, 15, 20, 25, 28,

$$\frac{15 + 20}{2} = \frac{35}{2} = 17.5$$

- The number having maximum frequency
- Mode = 3 median - 2mean, $34.1 = 3$ median - 2(20.6) ⇒ $34.1 = 3$ median - 41.2,

$$\text{Median} = \frac{75.3}{3} = 25.1$$

$$9. \frac{Q_3 + Q_1}{2}$$

$$10. \text{Mean} = np = \frac{n}{2}$$

$$\frac{n_{c1} + 2.n_{c2} + 3.n_{c3} + \dots + n.n_{cn}}{2^n},$$

$$\text{Variance} = \frac{n}{4}, \text{S.D.} = \frac{\sqrt{n}}{2}$$

$$11. \sigma^2 = \frac{\sum xi^2}{n} - \left(\frac{\sum xi}{n} \right)^2 =$$

$$\frac{4 + 16 + 36 + 64 + 100}{5} - \left(\frac{30}{2} \right)^2 = \frac{44}{5} - 36 = 8$$



LEVEL-II - (C.W)

MEAN (A.M, G.M, H.M)

1. The geometric mean of 10 observations on a certain variable was calculated as 16.2. It was later discovered that one of the observations was wrongly recorded as 12.9; infact it was 21.9. The correct geometric mean is

1) $\left(\frac{(16.2)^9 \times 21.9}{12.9}\right)^{1/10}$ 2) $\left(\frac{(16.2)^{10} \times 21.9}{12.9}\right)^{1/10}$
 3) $\left(\frac{(16.2)^{10} \times 12.9}{21.9}\right)^{1/10}$ 4) $\left(\frac{(16.2)^{11} \times 21.9}{12.9}\right)^{1/11}$

2. The A.M. of the observations 1.3.5, 3.5.7, 5.7.9,, $(2n-1)(2n+1)(2n+3)$ is ($\forall n \in N$)

1) $2n^3 + 6n^2 + 7n - 2$ 2) $n^3 + 8n^2 + 7n - 2$
 3) $2n^3 + 5n^2 + 6n - 1$ 4) $2n^3 + 8n^2 + 7n - 2$

3. The mean weight of 9 items is 15. If one more item is added tot he series the mean becomes 16. The value of 10th item is

1) 35 2) 30 3) 25 4) 20

4. The mean marks got by 300 students in the subject of statistics was 45. The mean of the top 100 of them was found to be 70 and the mean of the last 100 was known to be 20, then the mean of the remaining 100 students is

1) 45 2) 58 3) 68 4) 88

5. The average marks of boys in a class is 52 and that of girls is 42. The average marks of boys and girls combined is 50. The percentage of boys in the class is AIEEE - 2007

1) 60 2) 40 3) 20 4) 80

6. Mean of 'n' items is $\bar{x}$. If these n items are successively increased by 2, 2^2 , 2^3 ,, 2^n , then the new mean is

1) $\bar{x} + \frac{2^{n+1}}{n}$ 2) $\bar{x} + \frac{2^{n+1}}{n} - \frac{2}{n}$

3) $\bar{x} + \frac{2^n}{n}$ 4) $\bar{x} + 2^n$

7. The frequency distribution of discrete data given below, the frequency x against value 0 is missing.

Variable x : 0 1 2 3 4 5

Frequency f: x 20 40 40 20 4

If the mean is 2.5, then the missing frequency x will be _____

1) 0 2) 1 3) 3 4) 4

8. The minimum value of $(x-6)^2 + (x+3)^2 +$

$(x-8)^2 + (x+4)^2 + (x-3)^2$ is

1) 114 2) 141 3) 104 4) 2

9. Product of n positive numbers is unity. The sum of these numbers cannot be less than

1) 1 2) n 3) n^2 4) 2

10. An automobile driver travels from plane to hill station 100 km distance at an average speed of 30 km per hour. He then makes the return trip at average speed of 20 km per hour. What is his average speed over the entire distance (200 km)?

1) 25 km/hr 2) 24.6 km/hr
 3) 24 km/hr 4) 24.5 km/hr

11. If A.M. = 24.5, G.M. = 24.375 then H.M. =

1) 24 2) 24.125 3) 24.5 4) 24.25

MEDIAN & MODE

12. The minimum value of $|x-6| + |x+3| +$

$|x-8| + |x+4| + |x-3|$ is

1) 11 2) 21 3) 31 4) 42

13. If in a frequency distribution, the mean and median are 21 and 22 respectively, then its mode is approximately

1) 20.5 2) 22.0 3) 24.0 4) 25.5

M.D, S.D & VARIANCE

14. Mean deviation of the series a, a+d, a + 2d,-----, a + 2nd from its mean is

1) $\frac{(n+1)d}{(2n+1)}$ 2) $\frac{nd}{2n+1}$ 3) $\frac{(2n+1)d}{n(n+1)}$ 4) $\frac{n(n+1)d}{2n+1}$

15. If mean deviation through median is 15 and median is 450, then coefficient of mean deviation is

1) 1/30 2) 30 3) 15 4) 45

16. The mean and S.D. of 1, 2, 3, 4, 5, 6 is

1) 3, 3 2) $\frac{7}{2}, \sqrt{\frac{35}{12}}$ 3) $\frac{7}{2}, \sqrt{3}$ 4) $\frac{35}{12}$

17. If the S.D. of n observations $x_1, x_2, \dots, x_n$ is 4 and another set of n observations $y_1, y_2, \dots, y_n$ is 3 the S.D. of n observations $x_1 - y_1, x_2 - y_2, \dots, x_n - y_n$ is

1) 1 2) $2/\sqrt{3}$ 3) 5 4) 7

18. The variance of first 10 multiples of 3 is

1) 64.25 2) 54.25 3) 70.25 4) 74.25

19. Let r be the range and $S^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$.

If $S^2 \leq r^2 k$ then k is equal to

1) $\frac{1}{n-1}$ 2) $\frac{n}{n-1}$ 3) $\frac{n+1}{2(n-1)}$ 4) $\frac{1}{2(n-1)}$

20. The mean of the numbers $a, b, 8, 5, 10$ is 6 and the variance is 6.80, then which of the following gives possible values of a and b (AIEEE-2008)

1) $a = 0, b = 7$ 2) $a = 5, b = 2$
3) $a = 1, b = 6$ 4) $a = 3, b = 4$

21. Suppose a population A has 100 observations 101, 102, ..., 200 and another population B has 100 observations 151, 152, ..., 250. If V_A and V_B represent the variances of the two populations, respectively, then V_A / V_B is

1) 1 2) $9/4$ 3) $4/9$ 4) $2/3$

22. If the mean deviation about the median of the numbers $a, 2a, \dots, 50a$ is 50, then $|a|$ equal to

1) 4 2) 5 3) 2 4) 3

23. The variance of first 50 even natural numbers is

1) $\frac{833}{4}$ 2) 833 3) 437 4) $\frac{437}{4}$

24. All the students of a class performed poorly in Mathematics. The teacher decided to give grace marks of 10 to each of the students. Which of the following statistical measures will not change even after the grace marks were given? [JEE MAIN-2013]

1) mode 2) variance 3) mean 4) median

LEVEL-II (C.W) - KEY

1) 2 2) 4 3) 3 4) 1 5) 4 6) 2 7) 4
8) 1 9) 2 10) 3 11) 4 12) 2 13) 3 14) 4
15) 1 16) 2 17) 3 18) 4 19) 2 20) 4 21) 1
22) 1 23) 2 24) 2

LEVEL-II (C.W) - HINTS

$$1. \left(\frac{(16.2)^{10} \times 21.9}{12.9} \right)^{\frac{1}{10}}$$

$$2. \frac{\Sigma (2n-1)(2n+1)(2n+3)}{n} = \frac{\Sigma (4n^2-1)(2n+3)}{n}$$

$$= \frac{\Sigma (8n^3 + 12n^2 - 2n - 3)}{n}$$

$$= 2n^3 + 8n^2 + 7n - 2$$

$$3. 15 \times 9 = 135$$

$$16 \times 10 = 160$$

$$10\text{th item is } 160 - 135 = 25$$

$$4. \frac{\sum f_i \bar{x}_i}{\sum f_i}$$

$$5. \text{no. of boys} = b, \text{no. of girls} = g$$

$$(b \times 52) + (g \times 42) = (b + g) 50 \Rightarrow b : g = 4 : 1$$

$$6. \bar{X} + \frac{2(2^n - 1)}{n} = \bar{X} + \frac{2^{n+1}}{n} - \frac{2}{n}$$

$$7. \frac{\sum f_i x_i}{\sum f_i} = 2.5$$

$$8. \text{Minimum value obtained at the mean of } 6, -3, 8, -4, 3$$

$$9. A.M. \geq G.M$$

$$\frac{x_1 + x_2 + x_3 + \dots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \dots x_n},$$

$$\frac{x_1 + x_2 + \dots + x_n}{n} \geq 1 \Rightarrow \sum_{i=1}^n x_i \geq n$$

$$10. V(\text{Average}) = \frac{2V_1 V_2}{V_1 + V_2} = \frac{2 \times 30 \times 20}{30 + 20} = 24 \text{ km/hr}$$

$$11. G^2 = A.H$$

$$12. \text{Minimum value obtained at median of } -4, -3, 3, 6, 8$$

$$13. \text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$$

$$14. \text{Mean } \bar{x} = a + nd$$

$$M.D. = \frac{1}{(2n+1)} \sum |x_i - \bar{x}| = \frac{n(n+1)}{(2n+1)} d$$

$$15. \frac{M.D.}{\text{Median}}$$

16. $\bar{x} = \frac{\sum x_i}{n}, \sigma^2 = \left(\frac{n^2 - 1}{12}\right) d^2$
17. $V(aX + bY) = a^2 V(X) + b^2 V(Y)$
18. $\sigma^2 = \left(\frac{n^2 - 1}{12}\right) d^2$ where $n = 10, d = 3$
19. range ' r ' and variance ' σ^2 ' related by, $\sigma^2 \leq r^2$
- $$\frac{\sum (x_i - \bar{x})^2}{n} = \sigma^2 \Rightarrow \sum (x_i - \bar{x})^2 = n\sigma^2,$$
- $$S^2 = \frac{1}{(n-1)} \sum (x_i - \bar{x})^2, S^2 = \frac{n}{n-1} \sigma^2 \leq r^2 k$$
- $$\Rightarrow k = \frac{n}{(n-1)}$$
20. $\frac{a+b+8+5+10}{5} = 6 \Rightarrow a+b=7$
now use verification for variance
21. $V_B(x) = V_A(X+50) = V_A(X)$
22. $M.D = \frac{\sum |x_i - M|}{n}, M = \frac{51a}{2}$ and $n = 50 \Rightarrow |a| = 4$
23. $\bar{X} = \frac{\sum x_i}{n} = \frac{2+4+\dots+100}{50} = 51$
- $$\text{variance} = \frac{1}{n} \sum x_i^2 - (\bar{x})^2$$
- $$= \frac{1}{50} (2^2 + 4^2 + \dots + 100^2) - (51)^2 = 833$$
24. Median will go up by 2 and S.D. will remain same.



LEVEL-II - (H.W)

MEAN (A.M, G.M, H.M)

1. If a variable takes the values 0, 1, 2, n with frequencies proportional to the binomial coefficients ${}^n C_0, {}^n C_1, \dots, {}^n C_n$ then the mean of the distribution is
- 1) $\frac{n(n+1)}{4}$ 2) $\frac{n}{2}$ 3) $\frac{n(n-1)}{2}$ 4) $\frac{n(n+1)}{2}$

2. If a variable takes values 0, 1, 2, n with frequencies proportional to $q^n, {}^n C_1 p q^{n-1}, {}^n C_2 p^2 q^{n-2}, \dots, p^n$ where $p+q=1$ then the mean is
- 1) np 2) nq 3) npq 4) np^2
3. The A.M. of n observations is M, If the sum of (n-4) observations is a then the mean of remaining 4 observations is
- 1) $\frac{nM+a}{4}$ 2) $\frac{nM-a}{4}$ 3) $\frac{nM-a}{2}$ 4) $nM+a$
4. A distribution consists of three components with frequencies 300, 200 and 600 having their means 16, 8 and 4 respectively, then the mean of combined distribution is
- 1) 11 2) 10 3) 9 4) 8
5. When 10 is subtracted from all the observations, the mean is reduced to 60% of its value. If 5 is added to all the observations, then the mean will be
- 1) 25 2) 30 3) 60 4) 65
6. A student has obtained 75%, 80% and 85% in three subjects. If the marks of another subject are added then his average can not be less than
- 1) 60% 2) 65% 3) 80% 4) 90%
7. The mean weight of 150 students in a certain class is 60 kilograms. The mean weight of boys in the class is 70 kilograms and that of the girls is 55 kilograms, then the number of boys and girls are
- 1) 100, 50 2) 50, 100 3) 75, 75 4) 60, 90
8. The following table given, the average score of the students is
- | No. of students (f) | 8 | 12 | 20 | 10 | 6 | 4 |
|---------------------|----|----|----|----|----|----|
| Marks(x) | 20 | 30 | 40 | 50 | 60 | 70 |
- 1) 41 2) 42 3) 40 4) 39
9. The mean of following frequency table is 50.
- | Class | 0-20 | 20-40 | 40-60 | 60-80 | 80-100 | Total |
|-----------|------|-------|-------|-------|--------|-------|
| Frequency | 17 | f_1 | 32 | f_2 | 19 | 120 |
- The missing frequencies are
- 1) 28, 24 2) 24, 36 3) 36, 28 4) 28, 34
10. Let G_1, G_2 be the geometric means of two series $x_1, x_2, \dots, x_n; y_1, y_2, \dots, y_n$. If G is the geometric mean of $\frac{x_i}{y_i}, i = 1, 2, \dots, n$, then G is equal to
- 1) $G_1 - G_2$ 2) $\frac{\log G_1}{\log G_2}$ 3) $\frac{G_1}{G_2}$ 4) $G_1 + G_2$

11. If x_1, x_2, x_3 are three non zero real numbers such that $(x_1^2 + x_2^2)(x_2^2 + x_3^2) \leq (x_1x_2 + x_2x_3)^2$ then the G.M. of x_1, x_2, x_3 is

1) x_1 2) x_2 3) x_3 4) $\frac{x_1x_2x_3}{3}$

12. The harmonic mean of the numbers 2, 3, 4 is

1) 3 2) $\frac{1}{(24)^3}$ 3) $\frac{36}{13}$ 4) $\frac{13}{36}$

MEDIAN & MODE

13. If a variable taken discrete values $x+4, x-\frac{7}{2},$

$x-\frac{5}{2}, x-3, x-2, x+\frac{1}{2}, x-\frac{1}{2}, x+5$ ($x>0$)

then Median is

1) $x-\frac{5}{4}$ 2) $x-\frac{1}{2}$ 3) $x-2$ 4) $x+\frac{5}{4}$

14. The median wage of the worker in the following table is

Wages/ Week (Rs.)	No. of workers F	c.f
50 - 59	15	15
60 - 69	40	55
70 - 79	50	105
80 - 89	60	165
90 - 99	45	210
100 - 109	40	250
110 - 119	15	265

1) Rs. 80.08 2) Rs 82.08 3) 84.08 4) 81.04

15. The value of the mode given below is

Mark	0-10	10-20	20-30	30-40	40-50	50-60	60-70
Freq.	5	15	20	20	32	14	14

1) 43 2) 42 3) 41 4) 44

M.D, S.D & VARIANCE

16. Mean deviation of numbers 3, 4, 5, 6, 7 is

1) 0 2) 1.2 3) 5 4) 25

17. The mean deviation of the following distribution is

x	10	11	12	13	14
f	3	12	18	12	3

1) 12 2) 0.75 3) 1.25 4) 26

18. If $\sum_{i=1}^{18} (x_i - 8) = 9$ and $\sum_{i=1}^{18} (x_i - 8)^2 = 45$ then

the standard deviation of $x_1, x_2, \dots, x_{18}$ is

1) 4/9 2) 9/4 3) 3/2 4) 2/3

19. The mean square deviation of n observations $x_1, x_2, \dots, x_n$ about -2 and 2 are 18 and 10 respectively. Then S.D. of the given set is

1) 1 2) 2 3) 3 4) 4

20. The median and S.D. of a distribution are 20 and 4 respectively. If each item is increased by 2, the new median and S.D. are

1) 20, 6 2) 22, 6 3) 18, 6 4) 22, 4

21. The mean of five observations is 4 and their variance is 5.2. If three of these observations are 1, 2, and 6. Then the other two are

1) 2 and 9 2) 3 and 8 3) 4 and 7 4) 5 and 6

22. For two data sets, each of size 5, the variances are given to be 4 and 5 and the corresponding means are given to be 2 and 4, respectively. The variance of the combined data set is

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1) 5/2 2) 11/2 3) 6 4) 13/2

23. The sum of squares of deviations for 10 items from the mean ($=50$) is 250. The coefficient of variation is

1) 25 2) 50 3) 10 4) none

24. If the standard deviation of 10 observations $x_1, x_2, \dots, x_{10}$ is 4 and that of another set of 10 observations $y_1, y_2, \dots, y_{10}$ is 3 and also

$\bar{X}_i = (x_i - \bar{x})(y_i - \bar{y}), \bar{x}$ is mean of all x_i 's and

$\bar{y}$ is mean of all y_i 's. $\sum_{i=1}^{10} X_i = 80$ then standard

deviation of ten observations

$(x_1 - y_1), (x_2 - y_2), \dots, (x_{10} - y_{10})$ is

1) 1 2) 3 3) 5 4) $\sqrt{5}$

LEVEL-II (H.W) - KEY

1) 2 2) 1 3) 2 4) 4 5) 2 6) 1 7) 2
8) 1 9) 1 10) 3 11) 2 12) 3 13) 1 14) 3
15) 4 16) 2 17) 2 18) 3 19) 3 20) 4 21) 3
22) 2 23) 3 24) 2

LEVEL-II (H.W) - HINTS

1. Mean

$$= \frac{0.n_{c_0} + 1.n_{c_1} + 2.n_{c_2} + 3.n_{c_3} + \dots + n.n_{c_n}}{2^n}$$

$$= \frac{n.2^{n-1}}{2^n} = \frac{n}{2}$$

2. np, 3. $\frac{nM-a}{4}$, 4. $\frac{\sum f_i \bar{x}_i}{\sum f_i}$

5. $(\bar{x} - 10) = \frac{3}{5}\bar{x} \Rightarrow \bar{x} = 25$.

The mean required = $25 + 5 = 30$

6. Average cannot be less than the average of 75, 80, 85, 0

7. $(b \times 70) + (g \times 55) = (b + g)60 \Rightarrow b : g = 1 : 2$

8. $\frac{\sum f_i x_i}{\sum f_i}$

9. $f_1 + f_2 = 52$, use $\bar{x} = \frac{\sum f_i x_i}{N}$, $3f_1 + 7f_2 = 252$,

$N = \sum f_i$, $f_1 = 28, f_2 = 24$

10. $n\sqrt{x_1 x_2 x_3 \dots x_n} = G_1$,

$n\sqrt{y_1 y_2 \dots y_n} = G_2$,

$G = n\sqrt{\frac{x_1}{y_1} \cdot \frac{x_2}{y_2} \dots \frac{x_n}{y_n}} = \frac{G_1}{G_2}$

11. $(x_1^2 + x_2^2)(x_2^2 + x_3^2) \leq (x_1 x_2 + x_2 x_3)^2$ on expanding

$(x_1 x_3 - x_2^2)^2 \leq 0, \Rightarrow x_2^2 = x_1 x_3$

12. $H.M = \frac{3}{\frac{1}{2} + \frac{1}{3} + \frac{1}{4}} = \frac{36}{13}$

13. $x - \frac{7}{2}, x - 3, x - \frac{5}{2}, x - 2, x - \frac{1}{2}, x + \frac{1}{2}, x + 4, x + 5$

Median = $\frac{x - 2 + x - \frac{1}{2}}{2} = x - \frac{5}{4}$

14. Median = $l + \left(\frac{\frac{N}{2} - m}{f} \right) \times C$,

$l = 79.5, N = 265, m = 105, f = 60, c = 10$,
Median = 84.08

15. Mode = $l + \frac{(f_0 - f_1)}{2f_0 - (f_1 + f_2)} \times C$,

$l = 40, f_0 = 32, f_1 = 20, f_2 = 14, c = 10$,
Mode = 44

16. $\bar{x} = \frac{3 + 4 + 5 + 6 + 7}{5} = 5$,

Mean deviation (M.D) = $\frac{1}{n} \sum |x - x_i| = \frac{6}{5} = 1.2$

17. Mean deviation (M.D.) = $\frac{1}{N} \sum f_i |x_i - \bar{x}|$, where

$\bar{x} = A.M.$ $N = 48$, $\bar{X} = 12$, M.D. = 0.75

18. $\sigma^2 = \frac{1}{18} \sum_{i=1}^{18} (x_i - 8)^2 - \left[\frac{1}{18} \sum_{i=1}^{18} (x_i - 8) \right]^2$

$= \frac{1}{18} \cdot (45) - \left(\frac{1}{18} \cdot 9 \right)^2 = \frac{5}{2} - \frac{1}{4} = \frac{9}{4}$

$\therefore S.D. = \frac{3}{2}$

19. $\frac{1}{n} \sum (x_i + 2)^2 = 18$ ————— (1),

$\frac{1}{n} \sum (x_i - 2)^2 = 10$ ————— (2)

adding equation 1 & 2, $\frac{1}{2} \sum x_i^2 = 10$,

subtracting 2 from 1, $\frac{1}{n} \sum x_i = 1$, $\therefore S.D. = \sigma = 3$

20. Median is increased by 2 and
S.D does not change

STATEMENTS

$$21. \quad x + y = 11, \quad \frac{\sum x_i^2}{5} - 16 = 5.2, \quad \sum x_i^2 = 106, \\ x^2 + y^2 = 65, \quad x + y = 11, x^2 + y^2 = 65 \Rightarrow xy = 28, \\ \therefore x = 7, y = 4$$

$$22. \quad \sigma^2 = \frac{1}{n_1 + n_2} \left[n_1 \sigma_1^2 + n_2 \sigma_2^2 + \frac{n_1 n_2}{n_1 + n_2} (\bar{x}_1 - \bar{x}_2)^2 \right]$$

$$23. \quad \sum d^2 = 250, \quad \bar{x} = 50, \quad n = 10, \\ \sigma^2 = \frac{\sum d^2}{n} = 25 \Rightarrow \sigma = 5,$$

$$\text{Coefficient of variance} = 100 \times \frac{\sigma}{\bar{x}}$$

$$24. \quad \sigma_x = 4 \Rightarrow \sigma_x^2 = 16 \Rightarrow \sum \frac{(x_i - \bar{x})^2}{10} = 16$$

$$\sigma_y = 3 \Rightarrow \sigma_y^2 = 9 \Rightarrow \sum \frac{(y_i - \bar{y})^2}{10} = 9$$

$$\text{Let } Z_i = (x_i - y_i) \text{ Then } \sigma_z = \sqrt{\frac{\sum (z_i - \bar{z})^2}{n}}$$

$$= \sqrt{\frac{\sum [(x_i - \bar{x}) - (y_i - \bar{y})]^2}{10}}$$

$$= \sqrt{\frac{\sum (x_i - \bar{x})^2}{10} + \frac{\sum (y_i - \bar{y})^2}{10} - 2 \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{10}}$$

$$= \sqrt{16 + 9 - 2 \times \frac{80}{10}} = 3$$



LEVEL-III

ASCENDING & DESCENDING

1. I. The geometric mean of 2, 4, 16 and 32 is a
 II. The strength of 7 colleges in a city are 385, 1748, 1343, 1935, 786, 2874, 2108. Then the median strength is b .
 III. The algebraic sum of the deviations of 20 observations measured from 30 is 2. The mean of these observations is c .
 1) $a < b < c$ 2) $b < c < a$
 3) $c < a < b$ 4) $a < c < b$

2. Consider the following statements :
 i) Mean of 100 observations is 50 and standard deviation is 10. If 5 is added to each observation the new mean and standard deviation are 55, 10.
 ii) Mean of 100 observations is 50 and standard deviation is 10. If each observation is multiplied by 3 then the new mean and standard deviation are $50, \frac{10}{3}$.

The true statements are :

- 1) only (i) 2) only (ii)
 3) both (i), (ii) 4) neither (i) nor (ii)

MATCHING

3. Match the correct parts to make a valid statement :

List - I

List - II

A) Arithmetic Mean 1) $l + [f_2 / (f_1 + f_2)] \times i$

B) Geometric Mean 2) $(x_1 \cdot x_2 \cdot \dots \cdot x_n)^{1/n}$

C) Harmonic Mean 3) $\sum fX / \sum f$

D) Median 4) $l + \frac{N/2 - c \cdot f}{f} \times i$

E) Mode 5) $\left[\frac{1}{n} \left(\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n} \right) \right]^{-1}$

6) $l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times i$

A	B	C	D	E	A	B	C	D	E
1) 3	5	4	1	2	2) 2	4	1	5	3
3) 3	2	5	4	6	4) 1	3	2	4	6

4. A. Match the values of Q_1, Q_2 & Q_3 for the following data values
 13, 14, 7, 12, 17, 8, 10, 6, 15, 18, 21, 20

List - I

List - II

- A) Q_1 1) 13
 B) Q_2 2) 8.5
 C) Q_3 3) 17.5

A	B	C	A	B	C
1) 1	2	3	2) 2	1	3
3) 3	2	1	4) 3	1	2

ASSERTION & REASON

5. Statement-I : The variance of first n even

natural numbers is $\frac{n^2 - 1}{4}$

Statement-II : The sum of first n natural

numbers is $\frac{n(n+1)}{2}$ and the sum of squares

of first n natural numbers is $\frac{n(n+1)(2n+1)}{6}$

(AIEEE-2009)

1) Statement-I is true, Statement-II is true ; Statement-II is not a correct explanation for Statement-I

2) Statement-I is true, Statement-II is false

3) Statement-I is false, Statement-II is true

4) Statement-I is true, Statement-II is true, Statement-II is a correct explanation for Statement-I

6. Observe the following statements :

Statement-I : 10 is the mean of a set of 7 observations and 5 is the mean of a set of 3 observations. The mean of a combined set is 9.

Statement-II : If $\bar{x}_i (i = 1, 2, \dots, k)$ are the means of k -series of sizes $n_i (i = 1, 2, 3, \dots, k)$ respectively, then the combined or composite

mean is $\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2 + \dots + n_k \bar{x}_k}{n_1 + n_2 + \dots + n_k}$

1) Statement-I is true, Statement-II is true, Statement-II is a correct explanation for Statement-I

2) Statement-I is true, Statement-II is true; Statement-II is not a correct explanation for Statement-I

3) Statement-I is true, Statement-II is false

4) Statement-I is false, Statement-II is true

7. Let $x_1, x_2, \dots, x_n$ be n observations, and let $\bar{x}$ be their arithmetic mean and σ^2 be their variance,

Statement I: Variance of $2x_1, 2x_2, \dots, 2x_n$ is $4\sigma^2$

Statement II: Arithmetic mean of $2x_1, 2x_2, \dots, 2x_n$ is $4\bar{x}$ (AIEEE - 2012)

1) Statement-I is true, Statement-II is true, Statement-II is a correct explanation for Statement-I

2) Statement-I is true, Statement-II is true ; Statement-II is not a correct explanation for Statement-I

3) Statement-I is true, Statement-II is false

4) Statement-I is false, Statement-II is true

LEVEL-III - KEY

1) 4 2) 1 3) 3 4) 2 5) 3 6) 4 7) 3

LEVEL-III - HINTS

1. $a = (2.4.16.32)^{1/4} = 8$

$b = 1748$

$$c = A + \frac{\sum (x_i - A)}{n}$$

$$= 30 + \frac{2}{20} = 30.1$$

2. Mean of $ax_i + b = a\bar{x} + b$

$S.D. \text{ of } ax_i + b = |a| \sigma$

4. $Q_1 = \left(\frac{n+1}{4} \right)^{th} \text{ value}$

$$Q_2 = \frac{\left(\frac{n}{2} \right)^{th} \text{ value} + \left(\frac{n}{2} + 1 \right)^{th} \text{ value}}{2}$$

$$Q_3 = 3 \left(\frac{n+1}{4} \right)^{th} \text{ value}$$

7. $V(ax + b) = a^2 v(x) \quad \overline{2x} = 2\bar{x}$

